Problem 1. True or False

(a) (2 point) Binary attribute is not equivalent to discrete attribute.

True, the discrete attribute takes on finite or infinite number of distinct values, while the binary attribute is restricted by two values, ones and zeros.

(b) (2 points) For an unimodal curve that is not symmetric, according to the empirical formula (mean - mode) = -3(mean - median), its median and mode are always on the same side of its mean.

True, the median and mode are both either greater than the mean or less than. Therefore, the median and mode must be ont he same side of the mean

(c) (2 points) For a group of scalars, the median is always larger than its max and smaller than its min.

False, The median lies between the min and max values, meaning it cant be larger than the max or smaller than the min

(d) (2 points) Given two real-valued vectors, if their histogram is exactly the same, these two vectors are the same.

False, vectors can have the same histogram because histograms use value frequencies and not the order or exact position of elements

(e) (2 points) Given a similarity measure between two objects A and B, e.g., S(A, B), we can always define a valid distance measure as D(A, B) = 1/(S(A,B))

False, S(A,B) can be 0 which makes 1/(S(A,B)) undefined.

(f) (2 points) Given a vector of real numbers, if a raw number in the vector is smaller than the mean, its corresponding normalized value will be negative after z-score standardization.

True, Z-score standardization subtracts from the mean, so any value smaller than the mean becomes negative

(g) (2 points) Given any two vectors of the same length, the L2 distance (Euclidean distance) between them is always larger than the L1 distance (Manhattan distance) between them.

False, the L2 distance is never larger to the L1 distance and only less than or equal to

(h) (2 points) The goal of predictive (SUPERVISED) learning is to predict a single “target” or “outcome.”

True, the model is trained to try to predict the outcome using labeled data

(i) (2 points) The intelligence scale (IQ) is a ratio-scaled attribute.

False, IQ is an interval-scaled attribute because of how zero IQ does not mean no intelligence

(j) (2 points) The covariance matrix (of one random vector) has a square shape.

True, a covariance matrix is a vector with n components that's n x n so its always squared shape

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Problem 2. Data Measurement
Table 1: Heights and Weights of 12 NBA players

Player #	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	
Height (Cm)	195	200	192	230	212	200	210	205	218	177	185	
Weights (kg)	96	98	86	140	116	92	117	113	136	81	79	

(a) Compute the following statistical properties for both heights and weights.
(i) (2 points) Mean, mode, and median.

$$\text{Mean(Height)} = \frac{\text{sum of heights}}{12}$$

$$\text{Sum of Heights} = 195 + 200 + 192 + 230 + 212 + 200 + 210 + 205 + 218 + 177 + 185 + 197$$

$$\text{Sum of Heights} = 2421$$

$$Mean = \frac{2421}{12} = 201.75cm$$

Mode(Height) = 200cm (occurs twice)

$$\text{Sorted heights} = 177, 185, 192, 195, 197, 200, 200, 205, 210, 212, 218, 230$$

Median(Height) = 200 cm

$$\text{Mean(Weight)} = \frac{\text{sum of weights}}{12}$$

$$\text{Sum of Weights} = 96 + 98 + 86 + 140 + 116 + 92 + 117 + 113 + 136 + 81 + 79 + 92$$

$$\text{Sum of Weights} = 1246$$

$$\text{Mean(Weight)} = \frac{1246}{12} = 103.83 \text{ kg}$$

Mode(Weight) = 92 kg (occurs twice)

$$\text{Sorted weights} = 79, 81, 86, 92, 92, 96, 98, 113, 116, 117, 136, 140$$

$$\text{Median(Weight)} = \frac{96 + 98}{2} = 97 \text{ kg}$$

(ii) (2 points) First quartile, third quartile, and inter-quartile range.

$$\text{Sorted heights} = 177, 185, 192, 195, 197, 200, 200, 205, 210, 212, 218, 230$$

$$\text{Q1 (first quartile)} = \text{median of lower half} = \frac{192 + 195}{2} = 193.5 \text{ cm}$$

$$\text{Q3 (third quartile)} = \text{median of upper half} = \frac{210 + 212}{2} = 211 \text{ cm}$$

$$\text{IQR (interquartile range)} = Q3 - Q1 = 211 - 193.5 = 17.5 \text{ cm}$$

$$\text{Sorted weights} = 79, 81, 86, 92, 92, 96, 98, 113, 116, 117, 136, 140$$

$$\text{Q1 (first quartile)} = \text{median of lower half} = \frac{86 + 92}{2} = 89 \text{ kg}$$

$$\text{Q3 (third quartile)} = \text{median of upper half} = \frac{116 + 117}{2} = 116.5 \text{ kg}$$

$$\text{IQR (interquartile range)} = Q3 - Q1 = 116.5 - 89 = 27.5 \text{ kg}$$

(b) (5 points) Suppose we use z-score normalization to normalize the data for all these NBA players. After the normalization, what are the normalized heights and weights?

$$z = \frac{x - \mu}{\sigma}$$

Height (z-score)

$$\text{Player 1: } z = \frac{195 - 201.75}{16.77} = -0.40179$$

$$\text{Player 2: } z = \frac{200 - 201.75}{16.77} = -0.10449$$

$$\text{Player 3: } z = \frac{192 - 201.75}{16.77} = -0.57647$$

$$\text{Player 4: } z = \frac{230 - 201.75}{16.77} = 1.66740$$

$$\text{Player 5: } z = \frac{212 - 201.75}{16.77} = 0.60825$$

$$\text{Player 6: } z = \frac{200 - 201.75}{16.77} = -0.10449$$

$$\text{Player 7: } z = \frac{210 - 201.75}{16.77} = 0.48846$$

$$\text{Player 8: } z = \frac{205 - 201.75}{16.77} = 0.19286$$

$$\text{Player 9: } z = \frac{218 - 201.75}{16.77} = 0.96746$$

$$\text{Player 10: } z = \frac{177 - 201.75}{16.77} = -1.49344$$

$$\text{Player 11: } z = \frac{185 - 201.75}{16.77} = -0.99506$$

$$\text{Player 12: } z = \frac{197 - 201.75}{16.77} = -0.28487$$

Weights (z-score)

$$\text{Player 1: } z = \frac{96 - 103.83}{22.38} = -0.35055$$

$$\text{Player 2: } z = \frac{98 - 103.83}{22.38} = -0.26194$$

$$\text{Player 3: } z = \frac{86 - 103.83}{22.38} = -0.78760$$

$$\text{Player 4: } z = \frac{140 - 103.83}{22.38} = 1.60233$$

$$\text{Player 5: } z = \frac{116 - 103.83}{22.38} = 0.55469$$

$$\text{Player 6: } z = \frac{92 - 103.83}{22.38} = -0.53597$$

$$\text{Player 7: } z = \frac{117 - 103.83}{22.38} = 0.57918$$

$$\text{Player 8: } z = \frac{113 - 103.83}{22.38} = 0.40847$$

$$\text{Player 9: } z = \frac{136 - 103.83}{22.38} = 1.44255$$

$$\text{Player 10: } z = \frac{81 - 103.83}{22.38} = -1.02145$$

$$\text{Player 11: } z = \frac{79 - 103.83}{22.38} = -1.11373$$

$$\text{Player 12: } z = \frac{92 - 103.83}{22.38} = -0.53597$$

$$x' = 1 + \frac{x - x_{\min}}{x_{\max} - x_{\min}} \cdot 4$$

Height (min-max)

$$\text{Player 1: } x' = 1 + \frac{195 - 177}{53} \cdot 4 = 2.35849$$

$$\text{Player 2: } x' = 1 + \frac{200 - 177}{53} \cdot 4 = 2.94339$$

$$\text{Player 3: } x' = 1 + \frac{192 - 177}{53} \cdot 4 = 2.13208$$

$$\text{Player 4: } x' = 1 + \frac{230 - 177}{53} \cdot 4 = 5.00000$$

$$\text{Player 5: } x' = 1 + \frac{212 - 177}{53} \cdot 4 = 4.35849$$

$$\text{Player 6: } x' = 1 + \frac{200 - 177}{53} \cdot 4 = 2.94339$$

$$\text{Player 7: } x' = 1 + \frac{210 - 177}{53} \cdot 4 = 3.56604$$

$$\text{Player 8: } x' = 1 + \frac{205 - 177}{53} \cdot 4 = 3.15094$$

$$\text{Player 9: } x' = 1 + \frac{218 - 177}{53} \cdot 4 = 4.33962$$

$$\text{Player 10: } x' = 1 + \frac{177 - 177}{53} \cdot 4 = 1.00000$$

$$\text{Player 11: } x' = 1 + \frac{185 - 177}{53} \cdot 4 = 1.60377$$

$$\text{Player 12: } x' = 1 + \frac{197 - 177}{53} \cdot 4 = 2.50943$$

Weights (min-max)

$$\text{Player 1: } x' = 1 + \frac{96 - 79}{61} \cdot 4 = 2.11475$$

$$\text{Player 2: } x' = 1 + \frac{98 - 79}{61} \cdot 4 = 2.24590$$

$$\text{Player 3: } x' = 1 + \frac{86 - 79}{61} \cdot 4 = 1.45902$$

$$\text{Player 4: } x' = 1 + \frac{140 - 79}{61} \cdot 4 = 5.00000$$

$$\text{Player 5: } x' = 1 + \frac{116 - 79}{61} \cdot 4 = 3.45902$$

$$\text{Player 6: } x' = 1 + \frac{92 - 79}{61} \cdot 4 = 1.85147$$

$$\text{Player } 7\text{: } x' = 1 + \frac{117 - 79}{61} \cdot 4 = 3.57377$$

$$\text{Player } 8\text{: } x' = 1 + \frac{113 - 79}{61} \cdot 4 = 3.36065$$

$$\text{Player } 9\text{: } x' = 1 + \frac{136 - 79}{61} \cdot 4 = 4.54098$$

$$\text{Player } 10\text{: } x' = 1 + \frac{81 - 79}{61} \cdot 4 = 1.13114$$

$$\text{Player } 11\text{: } x' = 1 + \frac{79 - 79}{61} \cdot 4 = 1.00000$$

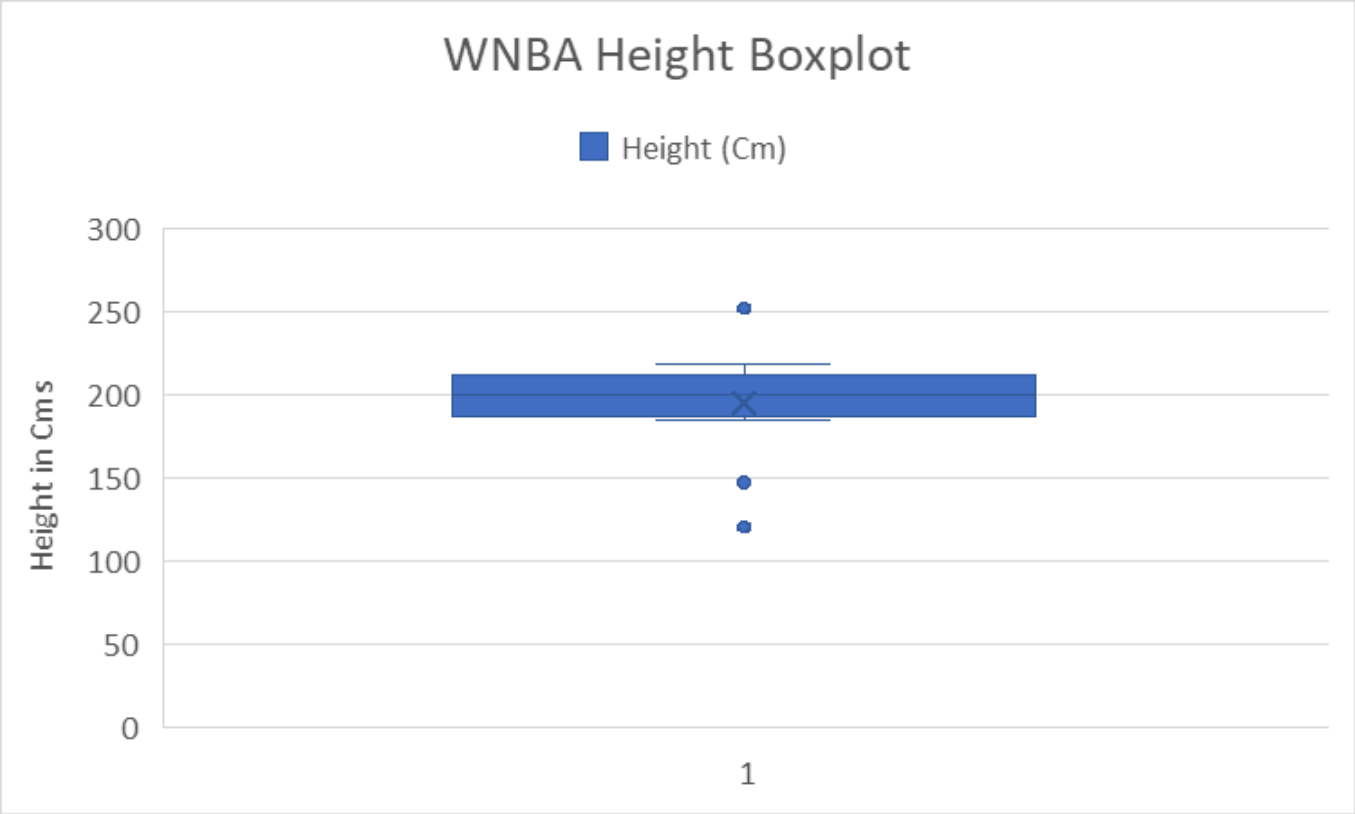
$$\text{Player } 12\text{: } x' = 1 + \frac{92 - 79}{61} \cdot 4 = 1.85147$$

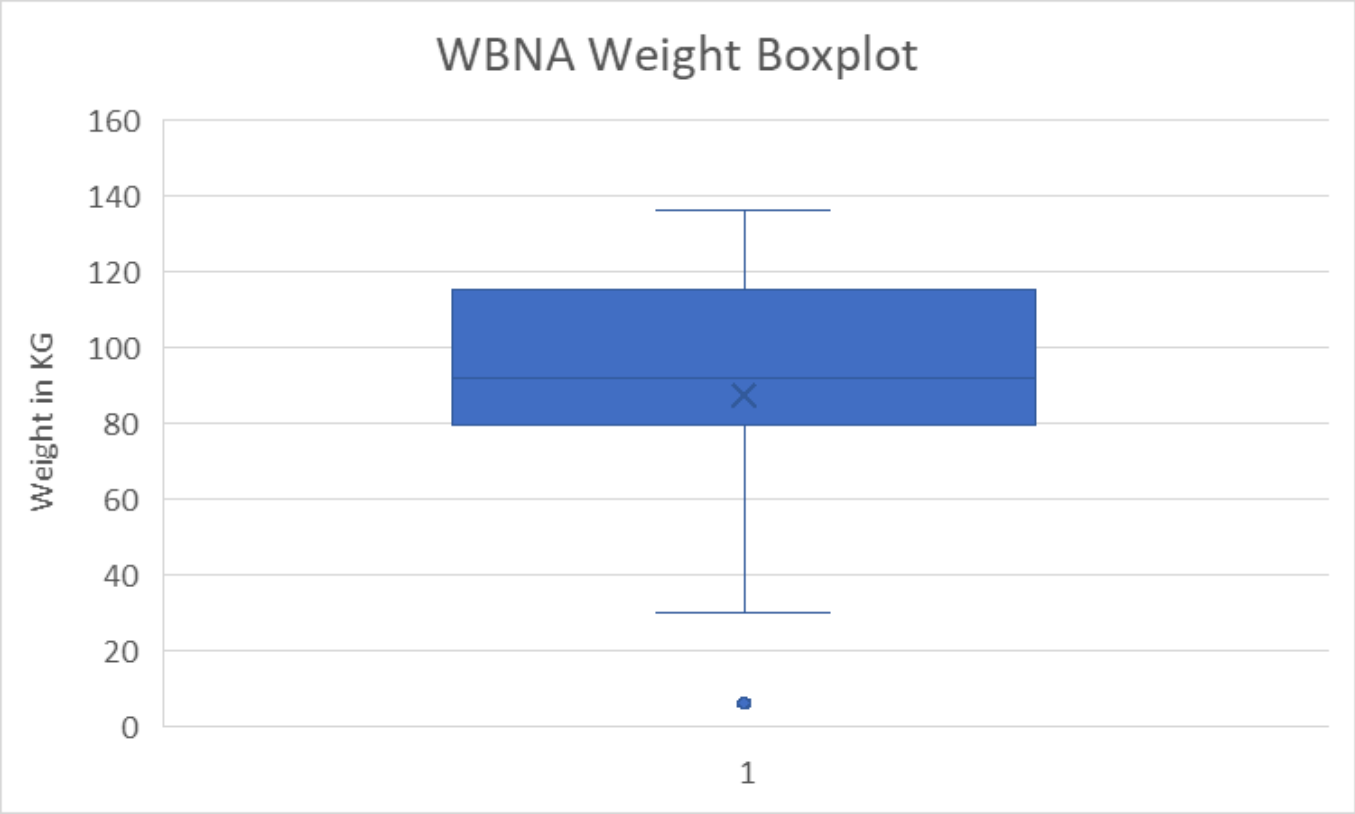
(c) (5 points) Suppose we use min-max normalization to normalize the data for all these NBA players to a new range (1,5) . After the normalization, what are the normalized heights and weights?

(d) (6 points) Plot the boxplots for both heights and weights of WNBA players in Table 2, and identify any outliers if they exist.

Table 2: Heights and Weights of 12 WNBA players

Player #	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**1**
Height (Cm)	120	200	192	252	212	200	210	205	218	147	185	1
Weights (kg)	6	98	86	30	116	92	117	113	136	81	79	9

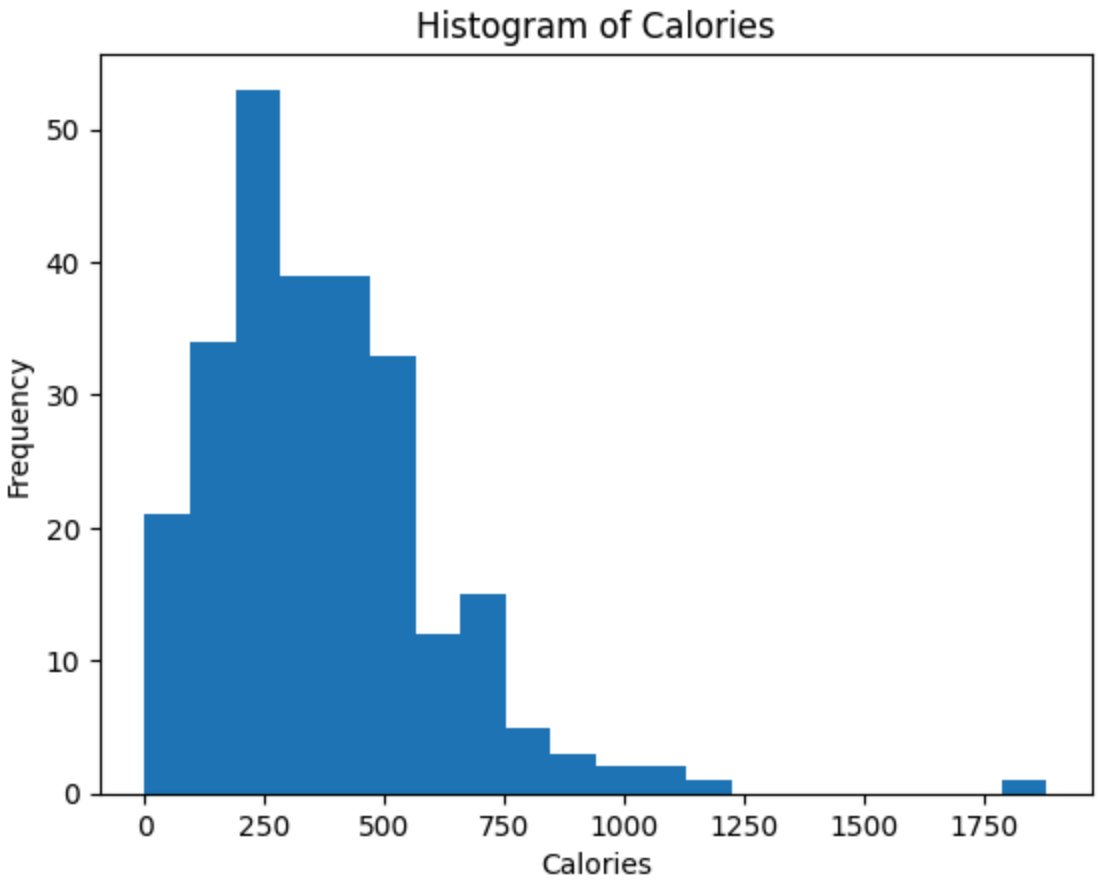




- Height outliers: Player 1 and Player 4
- Weight outliers: Player 1 and Player 4

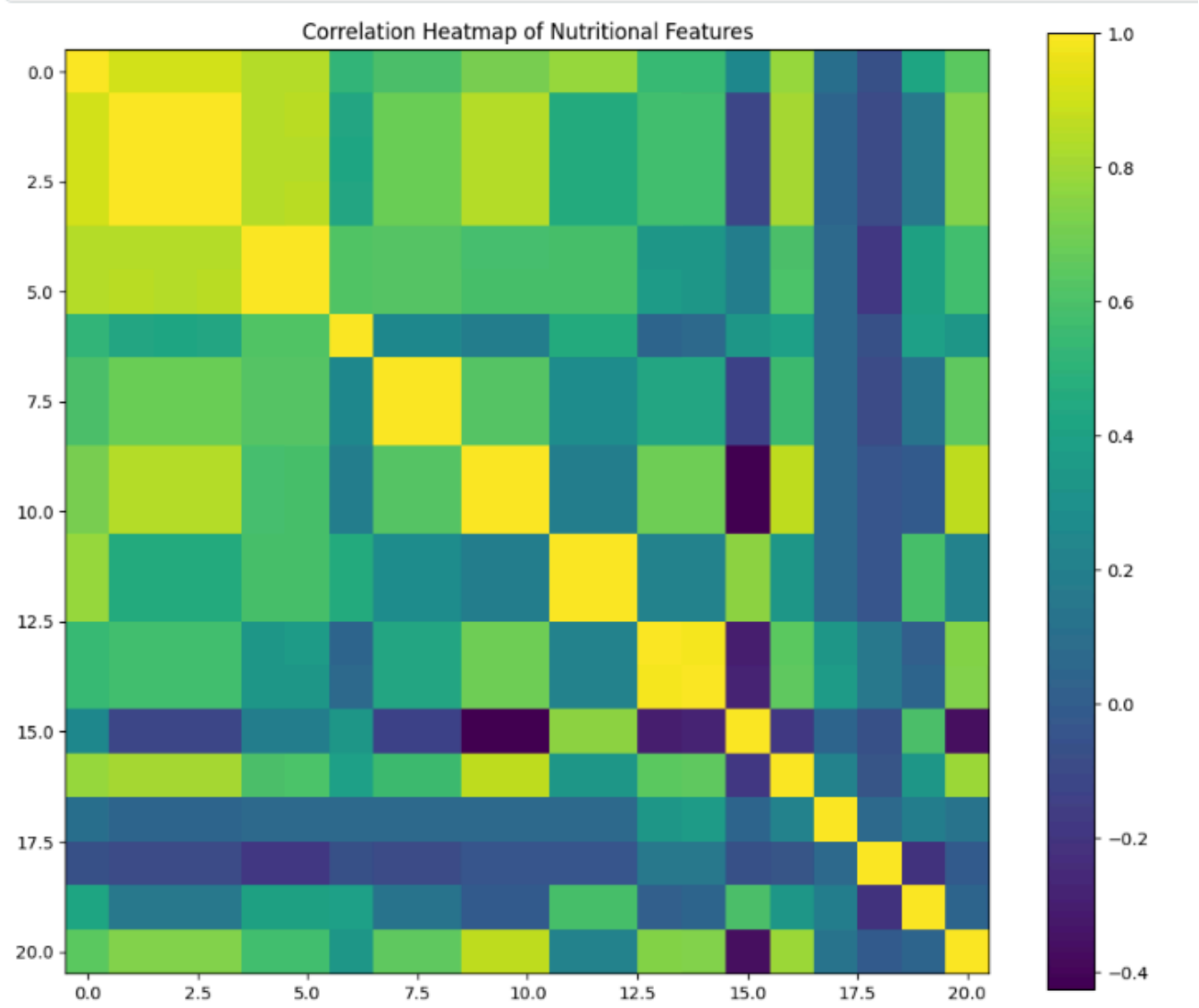
Problem 3. Data Exploration (40 Points)

(a) (6 points) Plot the histogram of the Calories. Comment on the datatype of the features



Calories is a numerical feature with an integer data type.

(b) (8 points) Plot the correlation heatmap among all features (Hint: the correlation matrix should be 21 × 21). You may notice that all diagonal entries are 1. Explain why this happens.

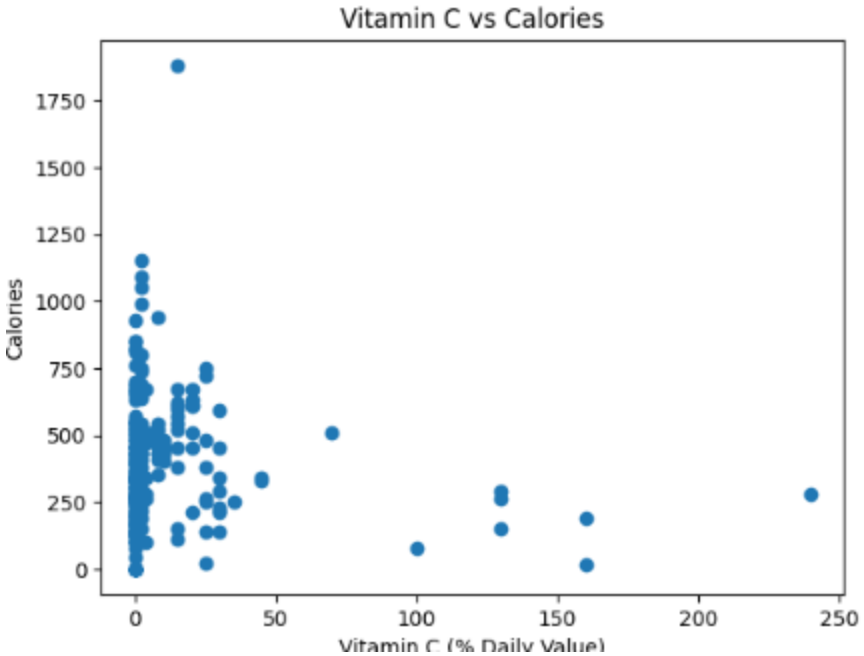


The diagonal entries are 1 because each feature is perfectly correlated with itself, resulting in a correlation coefficient of 1.

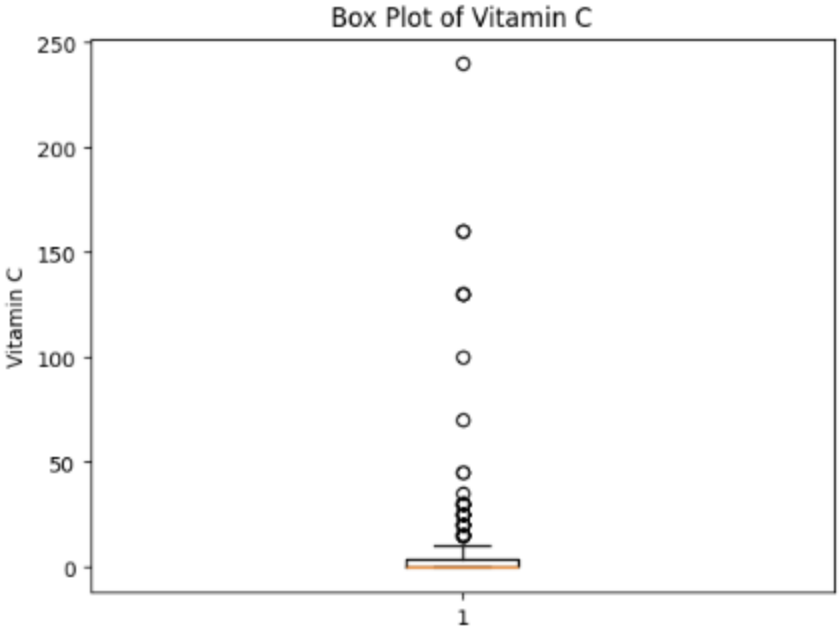
(c) (6 points) Report all features that have a negative correlation with Calories. Does this result match your expectations? Briefly explain why or why not.

Vitamin C has a negative correlation with Calories. This matches expectations because foods high in Vitamin C are often fruits or lighter items that are relatively low in calories.

(d) (10 points) Plot the scatter plot for “features vs. Calories” for all features found in (c).



(e) (10 points) Plot the box plot for all features found in (c).



Problem 4 Data Pre-processing: Missing Values

a) (6 points) Report the median and standard deviation for all numerical features

	Median	Std_Dev
Calories	340.0	240.269886
Calories from Fat	100.0	127.875914
Total Fat	11.0	14.205998
Total Fat (% Daily Value)	17.0	21.885199
Saturated Fat	5.0	5.321873
Saturated Fat (% Daily Value)	24.0	26.639209
Trans Fat	0.0	0.429133
Cholesterol	35.0	87.269257
Cholesterol (% Daily Value)	11.0	29.091653
Sodium	190.0	577.026323
Sodium (% Daily Value)	8.0	24.034954
Carbohydrates	44.0	28.252232
Carbohydrates (% Daily Value)	15.0	9.419544
Dietary Fiber	1.0	1.567717
Dietary Fiber (% Daily Value)	5.0	6.307057
Sugars	17.5	28.679797
Protein	12.0	11.426146
Vitamin A (% Daily Value)	8.0	24.366381
Vitamin C (% Daily Value)	0.0	26.345542
Calcium (% Daily Value)	20.0	17.019953
Iron (% Daily Value)	4.0	8.723263

(b) (8 points) Write the code to replace outliers of all numerical features in (a) with NaN (use the 3σ deviation to find outliers in slide week 2-2). Report the total number of NaN corresponding to each feature.

Calories	3
Calories from Fat	3
Total Fat	3
Total Fat (% Daily Value)	3
Saturated Fat	0
Saturated Fat (% Daily Value)	0
Trans Fat	9
Cholesterol	4
Cholesterol (% Daily Value)	4
Sodium	3
Sodium (% Daily Value)	3
Carbohydrates	5
Carbohydrates (% Daily Value)	5
Dietary Fiber	4
Dietary Fiber (% Daily Value)	4
Sugars	3
Protein	2
Vitamin A (% Daily Value)	6
Vitamin C (% Daily Value)	7
Calcium (% Daily Value)	0
Iron (% Daily Value)	3

(c) (6 points) Write the code to replace the missing values (NaN) with mean values. Report the median and standard deviation. Compare your result with (a) and write one sentence to explain your discovery.

Median values:

Calories	340.0
Calories from Fat	100.0
Total Fat	11.0
Total Fat (% Daily Value)	17.0
Saturated Fat	5.0
Saturated Fat (% Daily Value)	24.0
Trans Fat	0.0
Cholesterol	35.0
Cholesterol (% Daily Value)	11.0
Sodium	190.0
Sodium (% Daily Value)	8.0
Carbohydrates	44.0
Carbohydrates (% Daily Value)	15.0
Dietary Fiber	1.0
Dietary Fiber (% Daily Value)	5.0
Sugars	17.5
Protein	12.0
Vitamin A (% Daily Value)	8.0
Vitamin C (% Daily Value)	0.0
Calcium (% Daily Value)	20.0
Iron (% Daily Value)	4.0

```
Standard deviation values:
 Calories                          210.710032
Calories from Fat                  107.986184
Total Fat                           11.989233
Total Fat (% Daily Value)           18.452357
Saturated Fat                        5.321873
Saturated Fat (% Daily Value)       26.639209
Trans Fat                            0.330815
Cholesterol                         59.444949
Cholesterol (% Daily Value)         19.805418
Sodium                             520.229631
Sodium (% Daily Value)              21.676431
Carbohydrates                       25.225053
Carbohydrates (% Daily Value)        8.410548
Dietary Fiber                        1.416174
Dietary Fiber (% Daily Value)        5.705188
Sugars                              26.801587
Protein                             10.236827
Vitamin A (% Daily Value)           12.334957
Vitamin C (% Daily Value)            9.587148
Calcium (% Daily Value)             17.019953
Iron (% Daily Value)                 8.065942
```

It effected the standard deviation which measures how spread out the data because of how it replaced the outliers with the mean, the extreme values are removed making it less spread out.